

Optimization of Pearson and Blur

DV1674 | Assignment 2



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I. INTRODUCTION

II. METHOD

III. PEARSON

A simple bash script was written, `tests.sh`, with the purpose of making the testing and verification simpler. It has two modes: verify the output of the binary to previously established results and run the performance tests, which was used to extract the data. It ran all the tests on all the provided data files containing the various sizes of vectors, and later all threads. The parallel version of the binary is verified by the provided verification script, named `verify.sh`, making the verification part of `tests.sh` redundant and removed, only leaving the different performance tests.

III.I. STEPS

Compiler optimization

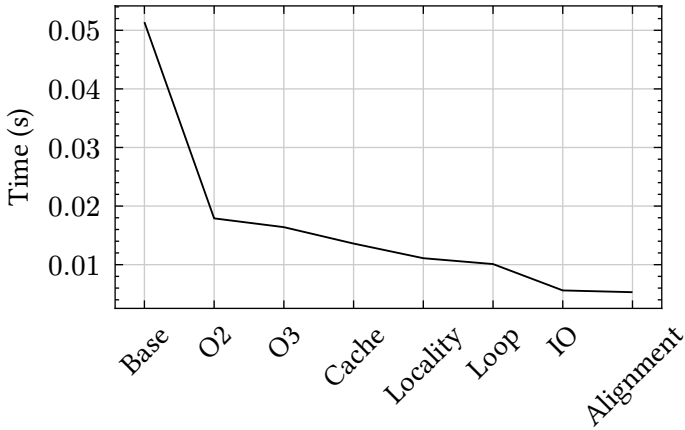
Cache implementation

Locality

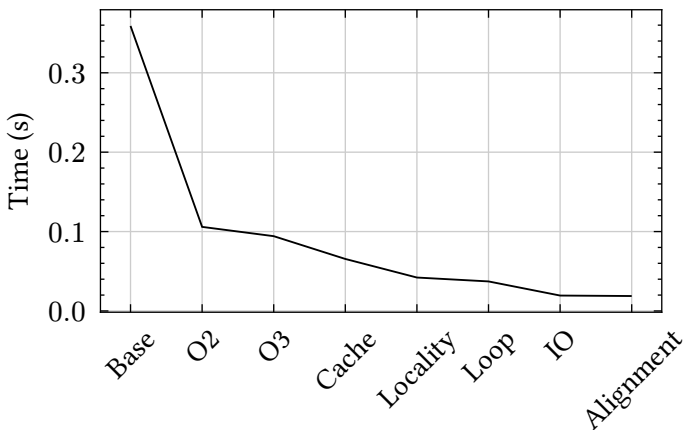
Loop unfolding

IO optimization

Vector alignment



Graph 1: Illustration of time per stage for size 128.

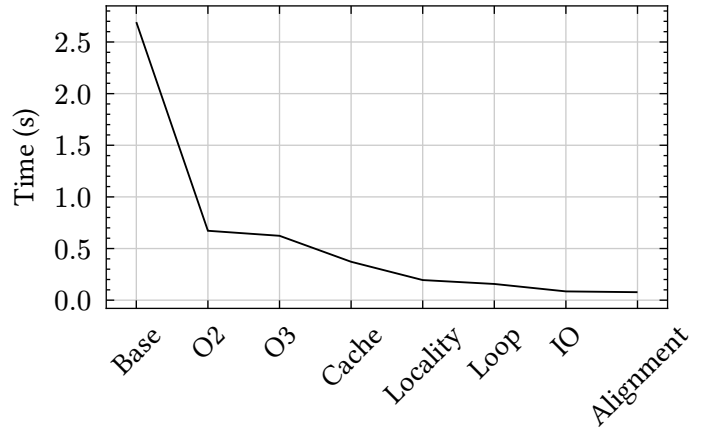


Graph 2: Illustration of time per stage for size 256.

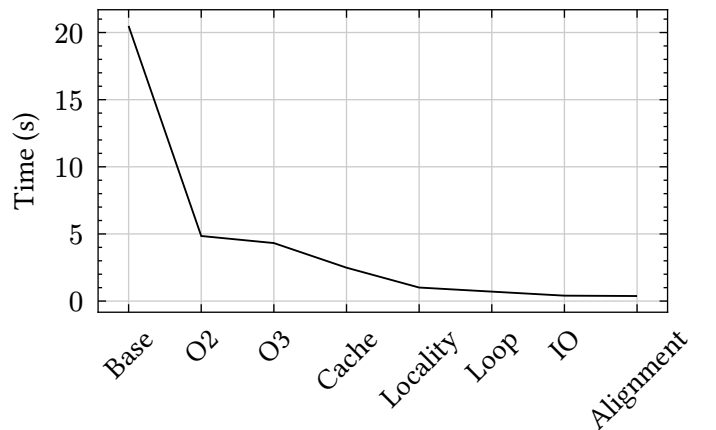
III.II. RESULTS

IV. BLUR

V. RESULTS



Graph 3: Illustration of time per stage for size 512.



Graph 4: Illustration of time per stage for size 1024.